PERSONAL / PUBLIC TECHNICAL PROJECT

TI C2000 actuator *controller.*

A F28069M control demonstrator organized around explicit command handling, state behavior, PWM output, setpoint validation and fault response.

TI C2000 F28069M SCI / UART EPWM

01 / OBJECTIVE

Make control *behavior explicit.*

The project documents a hardware-backed actuator controller on the TI C2000 F28069M. Its emphasis is on command flow, state transitions, output handling and observable fault behavior.

This is a public technical project. It is not presented as a certified safety product, client engagement or production deployment.

02 / SCOPE

Commands, *states, outputs.*

Technical scope

Firmware implements command handling and explicit control states with ePWM output and setpoint validation. Python tooling and documented firmware files support review of the behavior.

CONTROLLER

Texas Instruments C2000 F28069M.

EVIDENCE

Public source and project documentation.

STATE-MACHINE PLACEHOLDER

COMMAND → TIMEOUT → FAULT LATCH

Authorized state diagram or command table.

PWM EVIDENCE PLACEHOLDER

OUTPUT STATE CAPTURE

Scope capture or documented PWM behavior.

HARDWARE PLACEHOLDER

F28069M CONTROLLER PHOTOGRAPH

Authorized hardware image, with board and revision identified when available.

03 / EVIDENCE

Review the *implementation.*

Claims on this page are limited to the documented controller functions and technologies. Formal safety certification, product deployment and unverified reliability claims are intentionally excluded.

VIEW REPOSITORY ↗

RELATED SERVICE: EMBEDDED CONTROLS →

START A TECHNICAL DISCUSSION

Bring the *technical question.*

Describe the system, its current stage and the evidence available. Secure file exchange can follow the initial discussion.

START A TECHNICAL DISCUSSION →

Or write directly: info@herdersystemen.com

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